

GRADUATION PROJECT PROPOSAL

Academic Year 2026 – 2027

Smart Field Attendance & Workforce Verification System

with Integrated Shift Reporting & Fleet Accountability

A GPS-geofenced, on-device face recognition attendance platform with integrated shift reporting and shared-vehicle handover control, built for remote drilling and field-service operations.

Industry-Sponsored Project · Field Operations Digitization

Team of 7

Industry Partner: a water & gas well drilling company operating multiple field sites

Submitted to:

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1. EXECUTIVE SUMMARY

This proposal describes a single integrated mobile and web platform that digitizes three operational processes which our industry partner currently runs on paper: workforce attendance, site progress reporting, and shared-vehicle accountability.

The core of the system is a verified check-in: a worker can only register attendance when two independent conditions are satisfied at the same moment — his device is physically inside an approved site geofence, and a live capture of his face matches his enrolled biometric profile. Either signal alone is trivial to forge; combined with an active liveness check, they are not.

Layered on top of this verification core are two operational modules requested directly by the partner company. The first allows a site leader to define and activate a work site from the application itself, and to submit a structured progress report every twelve hours, replacing informal phone calls and messages with a timestamped, searchable record. The second is a fleet accountability module: because a single vehicle is driven by many different drivers, each vehicle carries a QR code that a driver must scan to claim it, followed by a mandatory photographic inspection of the vehicle before the trip begins and after it ends.

The entire platform is designed offline-first. Drilling sites are remote and frequently have weak or absent cellular coverage, so every record — attendance, report, and vehicle inspection — is captured and stored on the device immediately and synchronized automatically once connectivity returns.

Scope of this Proposal

- A statement of the operational problem as it exists today at the partner company.
- A description of the proposed solution and its four functional modules.
- The system architecture, data flow, and technology stack.
- A security and anti-fraud analysis of the system.
- Identification of the beneficiaries of the system.
- Expected project deliverables.

2. PROBLEM STATEMENT

Our industry partner drills water and gas wells across multiple remote and desert locations, frequently operating two or three active sites simultaneously. Workforce attendance, daily progress communication, and vehicle usage are all managed manually. Each of these creates a distinct operational and financial problem, and the problems compound as the number of active sites grows.

2.1 Attendance and Identity

- **No reliable identity verification.** Manual sign-in sheets can be completed by anyone on behalf of anyone else. There is no mechanism that binds an attendance record to the physical presence of a specific individual.
- **No proof of location.** There is no way to confirm that a worker was actually on the site he signed in for, or on any site at all.
- **Inaccurate working hours.** Shift start and end times, and consequently overtime, are estimated rather than measured. This produces recurring payroll disputes and prevents accurate labour cost control.
- **Site-blind records.** The same workforce rotates between two or three concurrent sites, but attendance is recorded company-wide rather than per site, so productivity cannot be attributed to a specific project.

2.2 Site Progress Reporting

- **Unstructured communication.** Site leaders currently report progress verbally, by telephone or messaging application. Reports are inconsistent in content, are not timestamped, and cannot be searched or compared across shifts.
- **No audit trail.** When a delay or an incident occurs, management has no reliable historical record of what was reported, by whom, and at what time.
- **Manual site definition.** A new drilling location cannot be brought into any tracking system without an administrator entering its coordinates manually from the office, which delays the start of monitored operations.

2.3 Shared Vehicle Fleet

- **No chain of custody.** Company vehicles are not permanently assigned. A single pickup or transport vehicle may be driven by five different drivers in one week, and no record exists of who held the vehicle at a given time.
- **Unattributable damage.** When damage is discovered, responsibility cannot be established, because there is no documented record of the vehicle's condition at each handover. Damage therefore goes unreported and uncharged.
- **Delayed maintenance reporting.** Mechanical faults are reported informally and late, increasing the risk of breakdown in remote desert conditions where recovery is expensive and slow.
- **Unverified distance.** Distance driven per trip is not recorded, so fuel consumption and vehicle utilization cannot be analysed.

2.4 Environmental Constraint

All of the above must be solved under a hard technical constraint: desert and remote drilling sites frequently have weak or no cellular signal. Any proposed digital solution that assumes continuous connectivity will fail in the field. Offline operation is therefore a requirement, not an optional feature.

3. PROPOSED SOLUTION

We propose a single platform consisting of a cross-platform mobile application used by workers, drivers, and site leaders, and a web-based administrative dashboard used by management. The platform is built around one principle: every operational event must be verified at the moment it happens, recorded with its context, and made impossible to construct after the fact.

3.1 Verification Principle

The system accepts an attendance check-in only when all of the following are true simultaneously:

1. The device reports a GPS position inside the geofence of a site to which the worker is assigned.
2. The operating system reports no mock-location provider in use.
3. An active liveness action (blink or head turn) is successfully performed.
4. The face embedding generated from the live capture matches the worker's enrolled embedding within a defined similarity threshold.

All four checks execute on the device itself, which is what allows the system to function with no network connection.

3.2 The Four Modules

Module	Purpose	Primary User
Module A — Workforce Verification	Geofenced, face-verified check-in and check-out with liveness detection and offline queuing.	Field worker
Module B — Site Management & Shift Reporting	In-app site creation by the site leader, and mandatory structured progress reports every twelve hours.	Site leader / supervisor
Module C — Fleet Accountability	QR-based vehicle claim, mandatory photographic inspection at handover, and trip logging.	Driver
Module D — Administrative Dashboard	Live attendance view, report archive, vehicle condition history, and anomaly review.	Management / HR

4. MODULE A — WORKFORCE VERIFICATION

This module preserves and refines the core attendance mechanism: proving that a specific person was physically present at a specific site at a specific time.

4.1 Enrollment

Each worker is enrolled once during onboarding. Several captures of his face are taken under varying conditions and converted into a numerical face embedding — a mathematical vector representation. The raw photographs are discarded; only the embedding is retained and linked to the worker identifier. This is both a privacy protection and a security measure, since an embedding cannot be reconstructed into a usable photograph.

4.2 Daily Check-In Flow

1. The worker opens the application at the start of his shift. No dedicated hardware or terminal is required.
2. The application reads the device GPS position and evaluates it against the geofences of all sites to which the worker is currently assigned.
3. If the worker is inside an active geofence, the application prompts a liveness action and captures the face.
4. An embedding is generated on-device and compared against the enrolled embedding.
5. The complete record — GPS coordinates, match result, liveness result, timestamp, and site identifier — is written to local storage immediately.
6. The record is uploaded automatically when connectivity returns, and the dashboard updates in near real time.

4.3 Check-Out and Hours Calculation

The same verified flow is repeated at the end of the shift. Worked hours are then calculated as the measured interval between two independently verified events, rather than being estimated. Overtime is derived from this measured interval against the shift definition configured for the site.

5. MODULE B — SITE MANAGEMENT & SHIFT REPORTING

This module is a direct response to two requests from the industry partner: that site definition should not require an office administrator, and that progress communication should move from informal messaging into the system itself.

5.1 In-App Site Creation by the Site Leader

A user holding the site leader role is granted the ability to define a work site directly from the mobile application, without depending on head office. Two methods are supported:

- **Capture current position.** The leader stands at the intended centre of the work area and taps "Use Current Location." The application records the device coordinates as the geofence centre.

- **Place a pin on the map.** For advance planning, the leader selects a point on an interactive map before travelling to the location.

In either case the leader then sets a radius appropriate to the physical extent of the drilling area, names the site, and assigns workers to it. The site becomes active immediately and workers assigned to it can begin checking in.

Governance note

Because the ability to create a geofence is also the ability to define where attendance is valid, site creation is restricted by role-based access control, every created or modified site is written to an immutable audit log with the identity of the creator and a timestamp, and management retains the ability to deactivate any site from the dashboard.

5.2 Twelve-Hour Shift Progress Report

The site leader is required to submit a structured progress report at the end of every twelve-hour shift cycle. The application issues a notification as the reporting window opens and escalates to management if the window closes without a submission.

Report contents

Field	Type	Purpose
Shift identifier	Automatic	Site, date, and day/night cycle, derived by the system.
Work completed	Structured text	Narrative of operations performed during the shift.
Depth / progress metric	Numeric	Quantitative progress figure appropriate to drilling operations.
Manpower on site	Automatic	Pulled directly from verified attendance records for the shift.
Equipment status	Selection + notes	Operational, degraded, or down, with explanatory text.
Incidents / HSE events	Yes / No + detail	Any safety event, injury, or near miss.
Delays and causes	Selection + notes	Weather, equipment, supply, or personnel related.
Photographs	Optional attachment	Visual evidence of progress or of a reported issue.
Voice note	Optional attachment	Practical alternative to typing in field conditions.

Each submitted report is timestamped, bound to the identity of the submitting leader, stored locally first, and synchronized when connectivity allows. On the dashboard, reports are presented as a chronological log per site, so management can review the full history of a project shift by shift.

Reports flagged with an incident or an equipment failure are surfaced separately for immediate attention.

6. MODULE C — FLEET ACCOUNTABILITY & DIGITAL HANDOVER

Company vehicles are a shared pool rather than individually assigned assets. This module establishes a digital chain of custody, so that at any moment the system knows which driver holds which vehicle, and in what condition that vehicle was received and returned.

6.1 Dynamic Vehicle Assignment

Every vehicle in the fleet carries a printed QR code fixed inside the cabin, encoding a unique vehicle identifier. To take a vehicle, a driver opens the application and scans this code, which opens a vehicle session recording the driver identifier, the vehicle identifier, and the start timestamp. The vehicle status changes from available to in use, and the system prevents a second driver from claiming the same vehicle concurrently. Releasing the vehicle closes the session and returns it to the available pool.

6.2 Mandatory Pre-Trip Inspection

The trip cannot begin until the driver completes a gated inspection sequence. The application will not enable the start-trip action while any required item is missing.

1. Four exterior photographs of the vehicle — front, rear, left side, and right side.
2. One photograph of the dashboard showing the current odometer reading and fuel level.
3. A confirmation prompt: are there any active warning lights or visible damages? If the driver answers yes, he must describe the condition in text or in a voice note before proceeding.

These records establish the documented condition of the vehicle at the moment the driver accepted it. Responsibility for any damage discovered subsequently can therefore be attributed to a specific custody period.

6.3 Mandatory Post-Trip Inspection

1. A final photograph of the odometer, from which the system calculates the exact distance driven during that driver's custody.
2. A mandatory checklist item: were any mechanical issues, unusual noises, or accidents experienced during this trip?
3. If the answer is yes, the driver records a voice note or types a description, which immediately raises an alert on the maintenance view of the administrative dashboard.

6.4 Trip Tracking

While a vehicle session is open, the application records the route travelled. Tracking begins at vehicle claim and stops at release, so location capture is scoped strictly to the period during which the driver holds a company vehicle. On release, the system stores a complete trip history: origin, route, destination, total distance, and duration. The administrative dashboard presents the live position of all vehicles with open sessions.

6.5 Open Technical Question — Location Source

The team will evaluate two architectures for vehicle location capture during the design phase and justify the selection in the final report.

Criterion	Driver mobile phone GPS	OBD2 hardware tracker
Cost	No additional hardware cost.	Per-unit device cost plus possible data subscription.
Reliability	Depends on the driver granting permissions, keeping the application running, and maintaining battery charge.	Independent of driver behaviour; powered by the vehicle.
Battery impact	Significant drain on the driver's personal device.	None.
Data available	Location only.	Location plus engine diagnostics and fault codes.
Integration effort	Background location APIs only.	Physical installation per vehicle and vendor API integration.

Our current assessment is that phone-based tracking is appropriate for the pilot phase, because it can be deployed immediately across the whole fleet at no hardware cost, while hardware trackers should be evaluated for production rollout, since they remove the dependency on driver compliance that otherwise undermines the accountability objective of this module.

7. SYSTEM ARCHITECTURE

7.1 Component Responsibilities

Layer	Responsibility	Key Technology
Mobile Application	Check-in flow, camera and GPS capture, liveness prompt, QR scanning, vehicle inspection capture, report forms, local storage and queue.	Cross-platform mobile framework with on-device ML runtime (TensorFlow Lite / ONNX Runtime).
Face Recognition	Embedding generation, matching against enrolled profiles, active liveness detection.	Pretrained backbone (FaceNet / ArcFace) fine-tuned and optimized for on-device inference.
Geofencing Logic	Point-in-radius evaluation against the worker's assigned active sites; site creation from device position or map pin.	Native GPS APIs, map SDK, mock-location detection.

Layer	Responsibility	Key Technology
Fleet & Session Engine	Vehicle session lifecycle, concurrency locking, inspection gating, odometer and distance computation, trip route storage.	QR decoding library, relational session model, background location service.
Reporting Engine	Twelve-hour report scheduling, notification and escalation, structured submission and attachment handling.	Scheduled notifications, media compression, local persistence.
Offline Sync Engine	Local persistence, retry and backoff queue, conflict handling, media upload management.	SQLite local database with a background sync worker.
Backend / API	Receives synced records, manages sites, workers, vehicles and roles, serves the dashboard.	REST API, relational database, containerized cloud deployment.
Administrative Dashboard	Live attendance and vehicle map, report archive, vehicle condition history, flagged-event review.	React single-page application with map integration.
Security Layer	Embedding-only biometric storage, encryption in transit and at rest, authentication and audit logging.	TLS, encrypted storage, role-based access control.

7.2 Data Flow

Stage 1 — Enrollment, performed once per worker

Face captured several times during onboarding, converted into a numerical embedding, and stored securely against the worker identifier. Raw images are not retained.

Stage 2 — Field capture, on-device and offline capable

Every field event — an attendance check-in, a shift report, or a vehicle inspection — is validated locally, written to the local database immediately with its full context, and queued for transmission. No event depends on network availability at the moment it occurs.

Stage 3 — Synchronization

Queued records upload through the API when connectivity returns. The server validates each record, stores it, and updates the dashboard in near real time. Anomalies — a detected mock location, a failed liveness attempt, a missed reporting window, or a reported vehicle fault — are surfaced to administrators as flagged events.

8. TECHNOLOGIES USED

Area	Technologies
Mobile Application	Cross-platform mobile framework; native GPS and camera APIs; QR code scanning library; background location services; local notification scheduling.
Artificial Intelligence	FaceNet or ArcFace face recognition backbone, fine-tuned for field conditions; active liveness detection module; TensorFlow Lite and ONNX Runtime for on-device inference and model optimization.
Local Storage & Sync	SQLite local database; background sync worker with retry and exponential backoff; media compression for photographs and voice notes.
Backend	REST API; PostgreSQL relational database; object storage for embeddings, inspection photographs, and report attachments; containerized cloud deployment.
Web Dashboard	React single-page application; map SDK for geofence setup, live vehicle tracking, and trip history rendering; role-based administrative views.
Security	TLS in transit; encryption at rest; embedding-only biometric storage; role-based access control; immutable audit logging.
Testing & QA	Automated unit and integration testing; face recognition accuracy evaluation across lighting, dust, and device-quality conditions; low-connectivity simulation; field usability testing with real workers.

9. SECURITY & TRUST

The value of the system depends entirely on the difficulty of falsifying the records it produces. The following table states each identified attack, the corresponding mitigation, and its implementation status within the scope of this project.

Attack or Risk	Mitigation
A worker checks in on behalf of an absent colleague.	Live face match required against the enrolled embedding.
A worker checks in from home using a photograph of himself.	Active liveness check (blink or head turn) required before capture is accepted.
GPS position falsified using a mock-location application.	Mock-location provider detection at the operating system level; suspicious positions flagged for review.

Attack or Risk	Mitigation
A site leader creates a fraudulent geofence around a convenient location.	Site creation restricted by role; every site creation and modification written to an immutable audit log; management can deactivate any site.
A driver takes a vehicle without accepting responsibility for its condition.	Trip cannot start until the full photographic inspection sequence is completed; vehicle session records driver identity and timestamp.
A driver photographs a different vehicle during inspection.	Inspection photographs bound to the scanned vehicle identifier, the device GPS position, and the capture timestamp.
Raw facial photographs leak if a device or the server is compromised.	Only mathematical embeddings are stored; raw images are discarded after enrollment.
GPS position falsified using specialised signal-level spoofing hardware.	Mock-location detection combined with pattern-based anomaly flagging on the dashboard for administrator review.
Checking in using a static photograph or a pre-recorded video of the worker.	Active liveness check requires a real-time action that a static image or simple replay cannot reproduce.

9.1 Privacy Considerations

- Biometric data is stored exclusively as non-reversible embeddings; no facial photograph is retained after enrollment.
- Vehicle location tracking is active only while a driver holds an open vehicle session, and stops on release. Drivers are not tracked outside working custody of a company vehicle.
- Worker consent for biometric enrollment will be planned and documented jointly with the industry partner during the requirements phase.

10. BENEFICIARIES

10.1 Primary Beneficiary — The Industry Partner

The direct beneficiary is our industry partner: a water and gas well drilling company operating multiple concurrent field sites, which is actively seeking a digital and verifiable solution to the problems described in Section 2. The company provides the operational requirements, access to field conditions for testing, and — where feasible — a live site for pilot deployment.

Value delivered to the partner

Business Function	Benefit
Payroll	Hours are measured rather than estimated, eliminating the principal cause of payroll disputes and reducing payment for hours not worked.
Labour cost control	Attendance is attributed to a specific site, enabling accurate per-project labour costing across concurrent operations.
Operational oversight	A structured, timestamped progress record every twelve hours replaces informal and unverifiable verbal reporting.
Asset protection	Documented vehicle condition at every handover makes damage attributable and therefore recoverable.
Maintenance	Faults are reported at the moment of discovery and routed directly to the maintenance function, reducing breakdown risk in remote areas.
Compliance & audit	A complete, immutable record of who was present, where, and when, available for internal and client audit.

10.2 Secondary Beneficiaries

- **Field workers.** Working hours and overtime are recorded objectively, protecting the worker against under-payment as much as it protects the company against over-payment.
- **Site leaders.** A site can be activated immediately from the field without waiting for head office, and reporting becomes a short structured form rather than a series of telephone calls.
- **Drivers.** A documented handover protects a driver from being held responsible for damage caused during another driver's custody.
- **Management and HR.** A single dashboard replaces paper sheets, scattered messages, and undocumented vehicle usage.

10.3 Wider Applicability

Although developed for a drilling company, the problem this system addresses — verifying that a specific person performed specific work at a specific remote location, and that shared assets were handled responsibly — is common to construction, surveying, agriculture, security services, logistics, and infrastructure maintenance. The architecture is therefore designed to be configurable rather than specific to a single industry.

11. EXPECTED DELIVERABLES

1. A functioning cross-platform mobile application supporting verified attendance, in-app site creation, twelve-hour shift reporting, and QR-based vehicle handover.
2. A web-based administrative dashboard providing live attendance, live vehicle positions, trip history, report archive, and flagged-event review.

3. Trained and optimized on-device face recognition and liveness models, with documented accuracy measurements across field conditions.
4. A documented offline-first synchronization protocol with measured behaviour under intermittent connectivity.
5. A written threat model with an explicit statement of defended and undefended attack surfaces.
6. A comparative evaluation of mobile-phone versus OBD2 hardware location architectures, with a justified recommendation.
7. A full technical report and pilot deployment findings from the industry partner's site.

A real problem, a real industry partner, and a real deployable system.

This project pairs a genuine machine learning component — on-device face recognition with liveness detection — with a substantial systems engineering challenge: reliable offline-first operation, verifiable accountability, and asset custody tracking across remote, low-connectivity field sites.